Performance Benefit from Dual-Frequency GNSS-based Aviation Applications under Ionospheric Scintillation: A New Approach to Fading Process Modeling

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ABSTRACT

Deep signal fading due to ionospheric scintillation may cause loss-of-lock on one or more satellites in GNSS receiver tracking loops, which can degrade the navigation availability of GNSS-based aviation applications. Scintillation impact can be mitigated via the frequency diversity, which decreases the chance of satellite loss in the presence of deep and frequent signal fades. This study presents an improved fading process model that generates correlated fading processes of dual-frequency signals and simulates the resulting scintillation impact to evaluate the availability benefit of utilizing dual-frequency GNSS in aviation applications under scintillation. The correlated fading process was described by combining single-frequency only and dual-frequency concurrent fading processes under the assumption that each fading process can be modeled as a Poisson process. Times between deep fading onsets and fading durations observed from GPS L1/L5 dual-frequency measurements collected at Hong Kong in March 2nd, 2014 were used to model the GPS L1 only, L5 only, and L1/L5 concurrent fading processes. Availability simulations for SBAS service supporting the LPV-200 phase of flight were conducted by considering the effects of satellite geometry degradation and shortened carrier smoothing time, which are caused by signal losses from deep fades generated by the newly proposed fading process model. A parametric analysis of availability resulting from variations in both the probability of loss-of-lock (under deep fading) and the receiver reacquisition time (following loss-of-lock) was conducted to provide receiver requirement standards for SBAS-based aviation under severe scintillation. The results show noticeable availability improvement from dual-frequency SBAS-based aviation applications over existing single-frequency SBAS.

1.0 INTRODUCTION

One of the most severe disruptions of Global Navigation Satellite System (GNSS) signals is caused by ionospheric scintillation. When electron density irregularities develop in the ionosphere, signals passing through the ionosphere suffer from rapid fluctuations in amplitude and phase which is referred to as ionospheric scintillation [1]. Especially in the equatorial region, small-scale irregularities known as plasma bubbles, arise and cause deep and frequent amplitude fading in GNSS signals [2]. When the signal amplitude fade below a certain threshold, known as deep fading, the receiver tracking performance deteriorates and can lead to signal loss-of-lock [3]. Especially when multiple satellite channels are lost simultaneously, the number of satellites available for use may decrease, and this degrades the navigation availability of GNSS-based aviation applications [4].

Previous studies have observed that the GNSS signal intensity during scintillation is decorrelated with frequency both in time and in magnitude, which implies that the impact of scintillation on GNSS-based aviation applications can be mitigated by using signals of different frequencies [5,6]. This is possible if the receiver maintains carrier tracking on at least one backup frequency signal, when it briefly loses a signal on another frequency due to scintillation. Despite this frequency diversity, the impact of scintillation still remains if different frequency signals experience deep fades at the same time because it brings a high risk of complete satellite loss. Hence, to analyze the impact of deep fades on GNSS-based aviation systems caused by scintillation, it is necessary to consider the correlation of deep fades within a fading process model. As part of the modernization of the Global Positioning System (GPS), a new navigation signal for civil use at L5 (1176.45 MHz, which is within the Aeronautical Radio Navigation Service (ARNS) spectrum), is now in service and is usable for future dual-frequency GNSS-based aviation applications in addition to the existing L1 (1575.42 MHz) signal [7].

A method to generate correlated fading processes of two channels was proposed in [8] to simulate the scintillation impact. Based on observing that the time between deep fades approximately follows an exponential distribution, the study modeled the number of deep fades over time as a Poisson process. A metric to measure the correlation level of two fading channels was proposed and applied to generate concurrent deep fades of two channels. This work was extended to generate continuous fading processes on GPS L1/L5 dual-frequency bands and to perform sensitivity studies based on the unknown correlation coefficient between L1 and L5 deep fades in time and the receiver reacquisition time following loss of lock caused by a deep fade [7].

Prior work in [7-9] was limited by certain assumptions that were made, including the rate of GPS L5 fading and the rate of L1/L5 concurrent deep fading, because L5 measurements were not available at that time. Moreover, deep fading was counted as a discrete event, i.e., a single point in time at which signal strength (as measured by C/N0) falls below a threshold. This discrete event based model only represents independent, instantaneous depletions and does not describe fading durations. This causes two potential problems. First, the probability of loss-of-lock for each fading event was assumed to be 100% regardless of fading duration. This results in a conservative availability assessment compared to that of allowing the probability of loss-of-lock to vary depending on the duration of each fading event. Second, the event of concurrent deep fading was defined based on the time interval between the local minima of two adjacent deep fades on different channels, which may lead to a conservative estimation of the degree of correlation between L1 and L5 deep fades.

This study proposes a new approach to model correlated fading processes by utilizing fading durations in addition to times between deep fading onsets. The duration of deep fades is applied to define concurrent deep fades by observing their overlaps along with the rate of deep fades. To reduce the conservatism of availability simulation, the probability of loss-of-lock for each fading event is specified by using fading duration information. A probability of loss-of-lock less than 100% was assumed for each time-step while the deep fading persists instead of applying a probability of 100% for each deep fading event. The duration and rate of deep fades are modeled by processing severe scintillation data collected at Hong Kong. Based on the new fading analysis and resulting fading process model, the availability benefit from using dual-frequency service was investigated in comparison with existing single-frequency service.

In Section 2.0, the dual frequency scintillation data used to analyze and extract fading characteristics is described. Section 3.0 explains the analysis method that was used to define deep fading and extract its characteristics. Section 4.0 describes the new correlated fading process model. Section 5.0 explains the availability simulation process and compares availability results of single and dual-frequency operations of Satellite Based Augmentation System (SBAS) for the Localizer Performance with Vertical guidance (LPV)-200 service under severe scintillation scenarios. Finally, conclusions are given in Section 6.0.

2.0 DATA

The data used in this study consists of 50 Hz GPS L1/L5 dual-frequency measurements collected using a Septentrio PolaRxS Pro scintillation receiver located in Hong Kong (22.3˚N, 114.2˚E). It is well known that strong scintillation events frequently occur in this equatorial region [2]. Figure 1 shows the scintillation index, S4, of the GPS L1/L5 dual-frequency bands on PRN 01 on March 2nd, 2014, especially during the severe scintillation period from 13:46:00 to 14:16:00 UTC [10]. Based on the observed scintillation data, times between deep fading onsets and fading durations are characterized in Section 3.0 and are utilized to construct a correlated fading process model in Section 4.0.

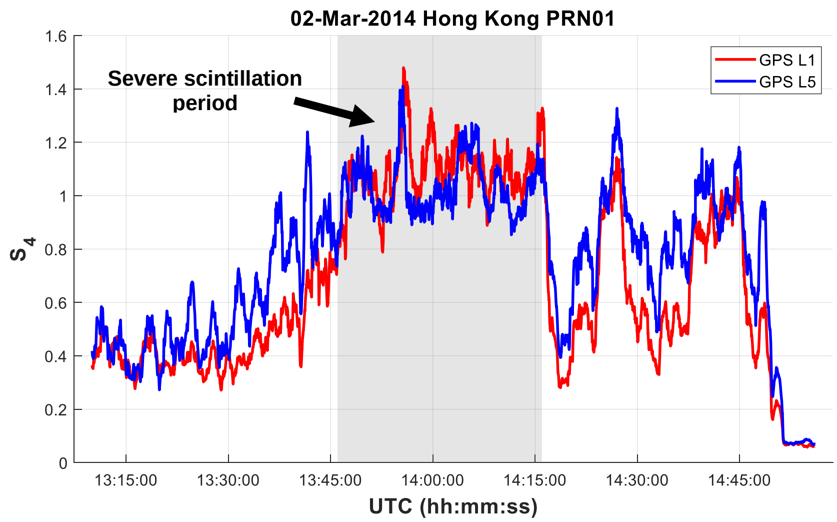

Figure 1. S4 index of GPS L1/L5 dual-frequency bands on PRN 01 on Mar 2nd, 2014, observed in Hong Kong.

3.0 SIGNAL FADING ANALYSIS

3.1. Definition of Deep Fading

Deep fading is defined as an event in which the signal intensity drops below a certain threshold which is potentially low enough to break the receiver's carrier tracking loop and cause loss-of-lock on the signal. The raw signal intensity is detrended using a moving average filter over a 60 second interval to remove low frequency contributions from satellite-receiver range variations, antenna patterns and background ionospheric/tropospheric delays. A threshold of -10 dB was used to define deep fading events within the detrended signal intensity, as shown in Figure 2 [6].

Two characteristics of deep fading, time between deep fading onsets and fading duration, were defined to analyze the impact of scintillation on aviation availability. The time between deep fading onsets is defined as the time interval between onsets of adjacent fading events whose signal intensity drops below the threshold. This parameter is used to measure the occurrence rate of deep fades in the presence of scintillation. The fading duration is defined as the duration of signal intensity dropping below and (later) rising above the threshold. It represents the time interval during which signal loss-of-lock may occur.

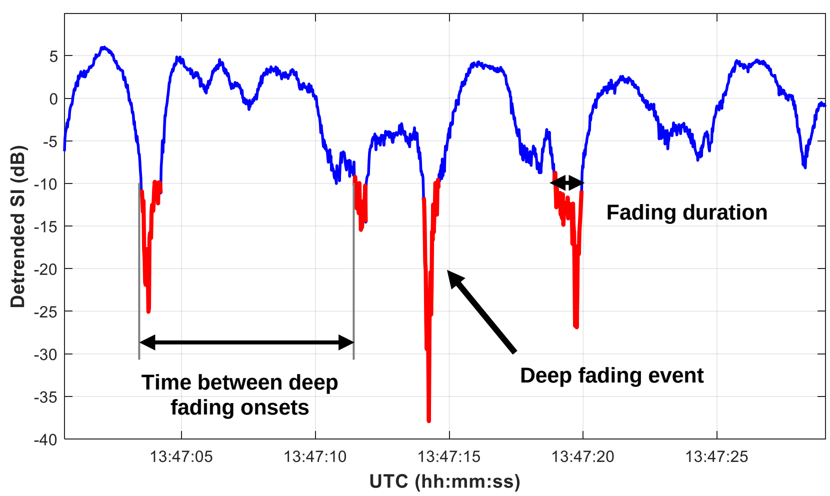

Figure 2. Illustration of detrended signal intensity and deep fading events with a threshold of -10 dB.

3.2. Concurrent Deep Fading

To describe the scintillation impact on dual-frequency GNSS aviation applications, it is necessary to analyze concurrent deep fading between two channels which may experience simultaneous signal outages. In previous work, concurrent deep fading was treated as occurring if discrete deep fades on different channels occurred within a constant time interval. A conservative time interval of longer than the 95th percentile of fading durations in the original data was used for this purpose [8]. However, this conservative approach generates more frequent concurrent deep fades than what is observed in real scintillation data.

To overcome this limitation, concurrent deep fading was determined based on actual overlaps between deep fades on different channels, which is made possible by modeling fading durations. As shown in Figure 3, GPS L1 only (L1 and not L5), L5 only (L5 and not L1), and L1/L5 concurrent deep fades are observed in the Hong Kong data. The duration of concurrent deep fading was defined as a union of overlapping GPS L1 and L5 deep fading events, which includes overlaps and the duration of L1 only or L5 only fades if they occurred consecutively. This was done because concurrent deep fading needs to be shared on both signals in order to maintain the assumption of a Poisson process. Figure 4 shows the distribution of times between concurrent deep fading onsets obtained from Hong Kong GPS L1/L5 dual-frequency measurements using the previous ([8]) and new approaches. The results show that the times between concurrent deep fading onsets determined from the new approach are longer than those of the previous approach. These less frequent concurrent deep fades, which represent real scintillation data more accurately, reduce the impact of scintillation on dual-frequency GNSS-based aviation availability due to simultaneous signal outages of both fading channels.

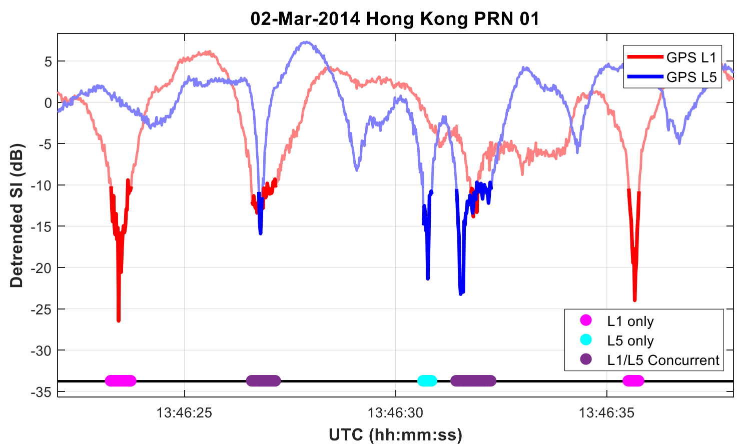

Figure 3. Signal fading examples across the GPS dual-frequency bands on PRN 01 on March 2nd, 2014, observed in Hong Kong. GPS L1 only, L5 only, and L1/L5 concurrent deep fades are flagged by determining the overlaps between L1 and L5 signal deep fades.

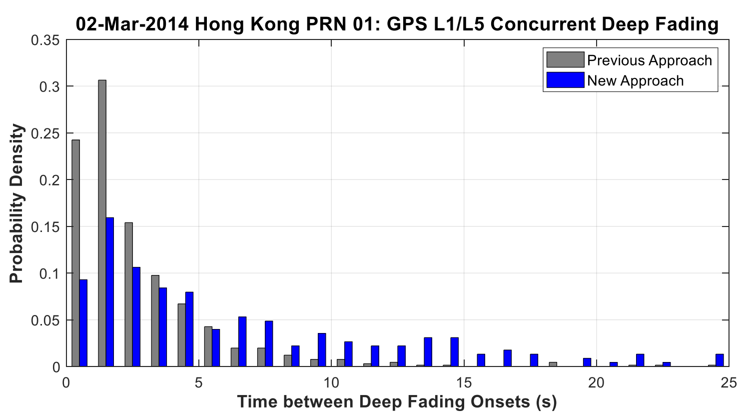

Figure 4. PDF of time between GPS L1/L5 concurrent deep fading onsets in the Hong Kong data. The concurrent deep fading onsets are determined by using the previous approach in [8] and the new approach presented in Section 3.2.

4.0 FADING PROCESS MODEL

4.1. Generation of Fading Process

To construct a model to generate correlated fading processes, empirical Probability Density Functions (PDFs) of the fading characteristics extracted from the Hong Kong data were obtained by normalizing their histograms. Figure 5 shows the probability densities of deep fading characteristics, time between deep fading onsets and fading duration, which follow their corresponding exponential distributions. The results indicate that more frequent deep fades occur in the GPS L5 signal (1176.45 MHz) than in the L1 signal (1575.42 MHz). This is consistent with the knowledge that lower frequency bands are more vulnerable to scintillation [6].

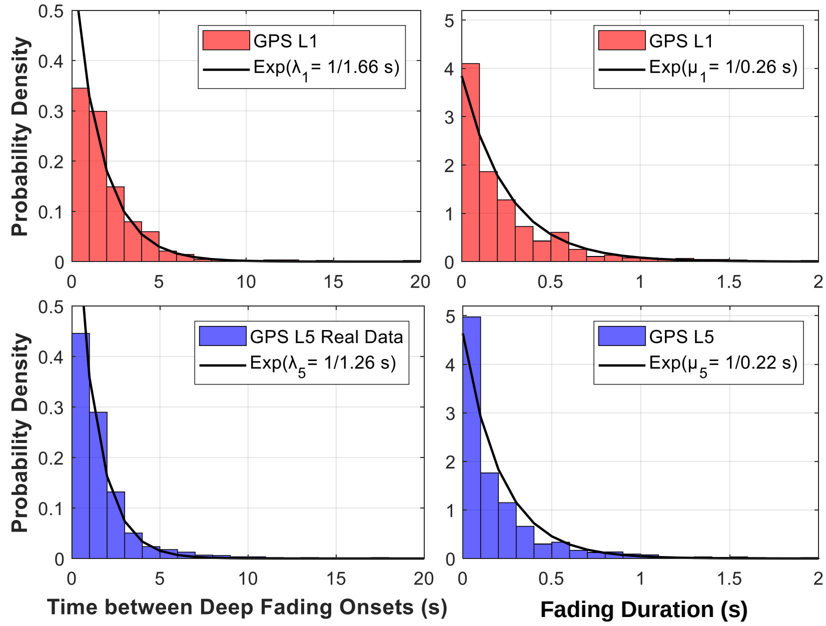

Figure 5. Histograms of time between deep fading onsets and fading duration for GPS L1 and L5 obtained from Hong Kong data. Each empirical PDF follows the exponential distribution with its corresponding rate.

Based on the observation that the time between deep fading onsets approximately follows an exponential distribution, the corresponding fading process was modeled as a Poisson process [8]. A Poisson process is a process of counting discrete events in which the arrival of a particular event is independent in time. One of the useful properties of a Poisson process is that corresponding inter-arrival times are exponentially distributed [11]. As shown in Figure 6, the time between deep fading onsets is modeled as an exponential distribution with a rate of λ, which is obtained from fits to the real scintillation data (see Figure 5). Additionally, the fading process model is improved by additionally modeling fading duration. It is also estimated from the actual scintillation data and is modeled by an exponential distribution with a rate μ. The resulting fading process can be simulated by generating exponentially distributed random variables with given rates λ and μ, respectively.

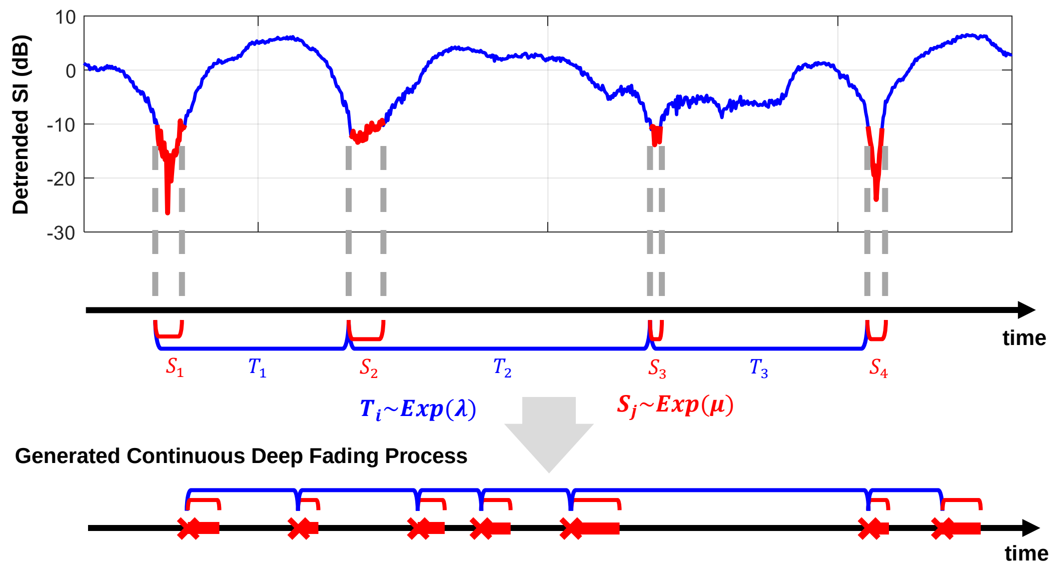

Figure 6. Modeling and Generating a fading process. Times between deep fading onsets, T_i and fading durations, S_j are modeled by corresponding exponential distributions with rates of λ and μ under the assumption of a Poisson process.

4.2. Correlated Fading Process Model

Another useful property of a Poisson process is that the combination of two independent Poisson processes of rates λ and λ' also follows a Poisson process with the summation of their rates, λ + λ' [11]. Using this property, a correlated fading process $N_1(t)$ (or $N_5(t)$) of rate λ_1 (or λ_5) for GPS L1 (or L5) can be modeled by combining a GPS L1 only (or L5 only) fading process $N_{1,only}(t)$ (or $N_{5,only}(t)$) of rate $\lambda_{1,only}$ (or $\lambda_{5,only}$) with a concurrent fading process C(t) of rate λ_{15}, as given in Equations (1) and (2) [7, 8].

$$\mathrm{N}_1[\lambda_1] = N_{1,only}[\lambda_{1,\mathrm{only}}] + C[\lambda_{15}] \tag{1}$$

$$\mathrm{N}_5[\lambda_5] = N_{5,only}[\lambda_{5,\mathrm{only}}] + C[\lambda_{15}] \tag{2}$$

Note that the fading process $N_1(t)$ has a rate of λ_1, which is the sum of $\lambda_{1,only}$ and λ_{15}, and the fading process $N_5(t)$ has a rate of λ_5, which is the sum of $\lambda_{5,only}$ and λ_{15}. For resulting fading processes $N_1(t)$ and $N_5(t)$ of given rates λ_1 and λ_5, the degree of correlation of deep fades between dual-frequency signals is determined by the concurrent fading process C(t) with a rate of λ_{15}.

Figure 7 shows PDFs of the fading characteristics of GPS L1 only, L5 only, and L1/L5 concurrent deep fades. Each PDF of the time between deep fading onsets is fitted into an exponential distribution with a corresponding rate by assuming that the fading process follows a Poisson process. The correlated fading processes of L1 (or L5) signal was obtained by combining an L1 only (or L5 only) fading process with an L1/L5 concurrent fading process. An interesting observation is that the resulting fading process of L1 has a rate of $\lambda_{1,only} + \lambda_{15}$ = 1/2.19 (s) + 1/7.13 (s) = 1/1.68 (s). This is almost the same as the empirically obtained rate of L1 fading process, λ_1 = 1/1.66 (s), as shown in Figure 5. Similarly, the combined fading process of L5 has a rate of $\lambda_{5,only} + \lambda_{15}$ = 1/1.50 (s) + 1/7.13 (s) = 1/1.24 (s), which is also very close to the empirically obtained rate of L5 fading process, λ_5 = 1/1.26 (s). This observation shows that the fading process model works well under the assumption of Poisson-process behavior.

Each PDF of the fading duration was also fitted into an exponential distribution with a corresponding rate, respectively. Fading durations were independently generated after each deep fading onset for each fading process. Figure 7 shows that fading durations of L1/L5 concurrent deep fades are distributed with much larger mean value than that of L1 only and L5 only deep fades. This is because the duration of concurrent deep fading is defined as a union of overlapped L1/L5 deep fades and consecutive L1 only or L5 only deep fades. With this definition, the concurrent deep fading process C(t) is treated as independent events shared by both L1 and L5 correlated fading processes, $N_1(t)$ and $N_5(t)$.

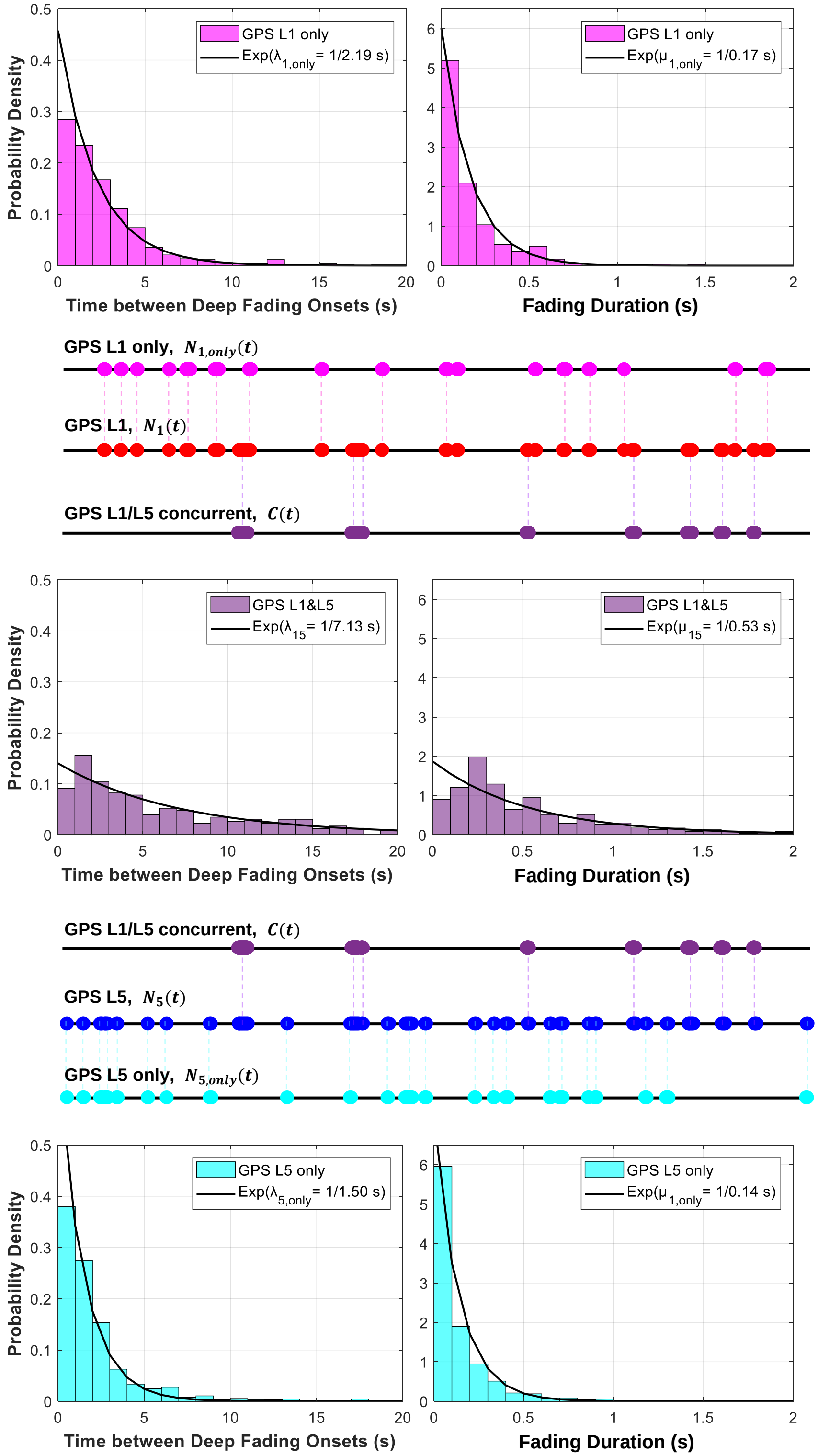

Figure 7. Generation of GPS L1 and L5 correlated fading processes from independent L1 only, L5 only, and L1/L5 concurrent fading processes. Each fading process is fitted into an exponential distribution with its corresponding rate.

5.0 AVAILABITY SIMULATION

5.1. Simulation Specifications

Based on the improved fading process model given in Section 4.0, this section provides an assessment of the availability of GNSS-based aviation applications under severe scintillation. This study assumed the use of SBAS supporting the LPV-200 phase of flight with a Vertical Alert Limit (VAL) of 35 m. Vertical Protection Levels (VPLs) for a single user at Hong Kong (22.3˚N, 114.2˚E) were calculated and compared to VAL at every second during the 6-hour period of severe scintillation observed in the data from Hong Kong on March 2nd, 2014. These simulations used the GPS constellation with a visibility mask angle of 5˚ during this period. A 1 m User Range Accuracy (URA) is assumed for satellite clock/ephemeris error. The troposphere error model and airborne code noise/multipath model from the WAAS Minimum Operational Performance Standards (MOPS) were also used. In the case of dual-frequency service, the iono-free dual-frequency code noise and multipath model was used [12]. The residual error for ionospheric corrections in the single-frequency service was based on a Grid Ionospheric Vertical Error Indicator (GIVEI) index of 11 (4.5 m, 99.9%). When one frequency is lost in dual-frequency service, the GIVEI of 11 used in the single-frequency service was also used to support temporary single-frequency operations.

These availability simulations were conducted under severe scintillation scenarios which are much worse than the observed Hong Kong data. In these scenarios, all satellite channels in view were assumed to experience severe deep and frequent fades generated by the correlated fading process model introduced in Section 4.0. Possible signal outages were assumed to occur at every epoch of simulated deep fades with a sampling rate of 50 Hz according to the two parameters: probability of loss-of-lock (under deep fading) and receiver reacquisition time (following loss-of-lock). The chance of signal loss-of-lock for each fading event was specified by assuming an independent chance of loss-of-lock for each 50 Hz interval while deep fade persists. Since the probability of loss-of-lock and the reacquisition time depend on the receiver tracking performance, both of these parameters were parametrically analyzed to cover a range of possible cases of receiver performance. Figure 8 shows an example of the generation of correlated deep fading events and resulting loss-of-lock events that feeds the availability simulation.

The Matlab Algorithm Availability Simulation Tool (MAAST) [13] was modified to incorporate scintillation impact generated from the correlated fading process model and resulting signal loss-of-lock events. Scintillation impact on the aviation availability were considered in two ways. At first, loss-of-lock due to deep fading is considered which can lead to satellite geometry degradation during outage periods, since lost satellite channels cannot be used for position calculations. Another possible scintillation impact is due to shortened carrier smoothing times, which leads to higher code noise levels. Most GNSS-based aviation applications use a Hatch filter to lower the noise level of pseudorange measurements. However, frequent losses-of-lock reduce the effective smoothing time, leading to consistently high noise levels on pseudorange measurements. The relative noise level on pseudorange measurements with respect to the fully converged value is modeled as $1 + 9e^{-\Delta t/100}$, where Δt is the time after filter reset (which starts once a lost satellite is reacquired) [9]. The time constant of the Hatch filter was set at 100 s as specified in the WAAS MOPS [12].

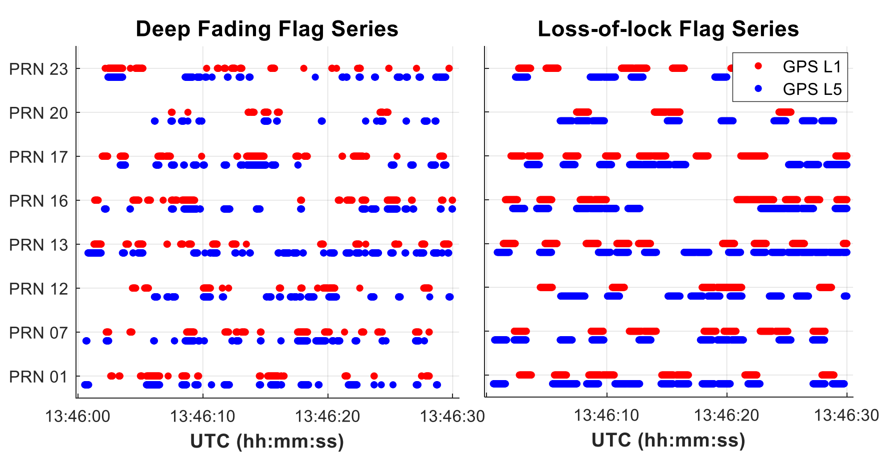

Figure 8. Examples of correlated fading processes and resulting loss-of-lock events under the severe scintillation scenario (Probability of loss-of-lock of 10% and reacquisition time of 1 s).

5.2. Availability Results

Figure 9 shows availability results for the LPV-200 phase of flight under the severe scintillation scenarios described in Section 5.1. The simulations were parametrically analyzed based on the two parameters mentioned above, probability of loss-of-lock and reacquisition time, which depend on receiver tracking performance. As shown in Figure 9, both parameters significantly affect the availability results. A higher probability of loss-of-lock increases the chance of signal losses during the duration of deep fading, while a longer reacquisition time increases the chance of simultaneous loss of satellites. The results show significant availability improvement by using dual-frequency SBAS compared to single-frequency SBAS for all combinations of these receiver tracking parameters. For example, a single-frequency service needs immediate reacquisition of receiver signal tracking to meet 95% availability when the probability of loss-of-lock is 15%, while for a dual-frequency service, reacquisition within 1 second is sufficient to provide more than 95% availability.

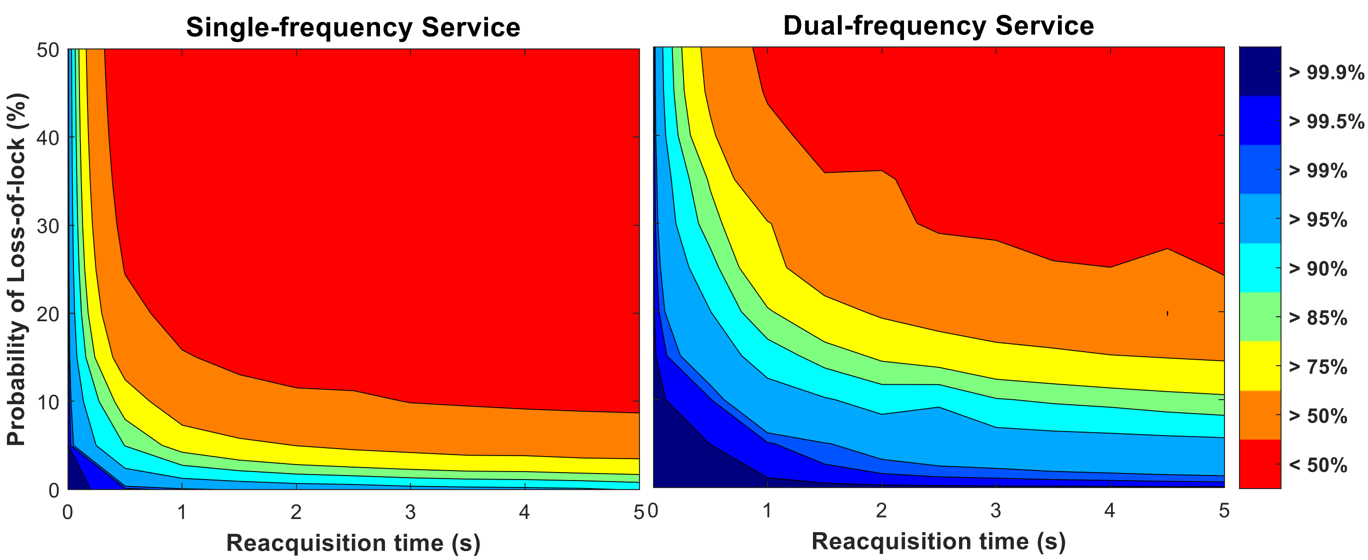

Figure 9. LPV-200 availability results under severe scintillation scenarios for the aviation operations of GPS L1 single-frequency and L1/L5 dual-frequency services.

Availability analysis while varying the GIVEI was also conducted to investigate the sensitivity of availability to ionospheric correction error for single-frequency SBAS. Ionospheric correction error is likely to increase under significant ionospheric scintillation because electron density irregularities are likely to degrade the spatial consistency of ionospheric delay and thus the quality of delay estimation by SBAS ground station networks. As shown in Figure 10, this sensitivity study shows that the availability benefit from dual-frequency service increases as ionospheric correction errors increase under severe scintillation scenarios. This is not surprising because dual-frequency SBAS directly corrects for ionospheric delays and is not sensitive to GIVEI except when loss of a satellite on one of the two frequencies causes a temporary downgrade to single-frequency SBAS for that satellite.

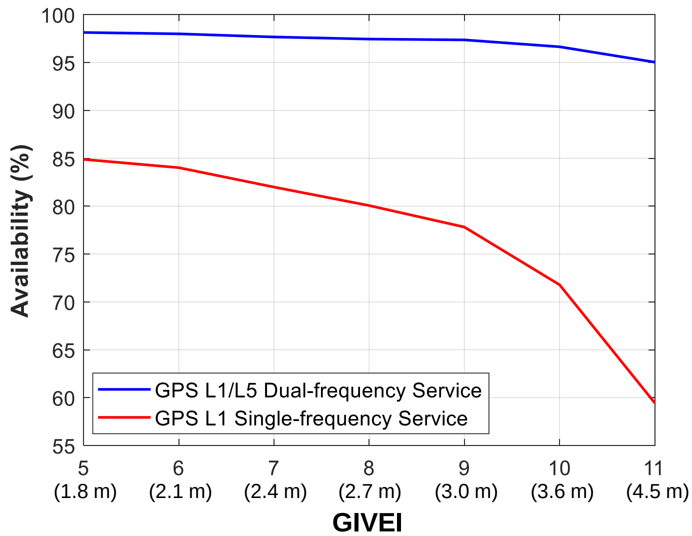

Figure 10. Availability assessment of GPS L1 single and L1/L5 dual-frequency services based on varying GIVEI. Probability of loss-of-lock and reacquisition time are assumed to be 15% and 1 s, respectively.

6.0 CONCLUSION

To assess GNSS-based aviation availability under scintillation, *Seo et al.* [7-9] proposed a way to generate correlated fading processes and evaluated the availability of dual-frequency GNSS-based aviation applications. However, it did not consider fading characteristics obtained from dual-frequency measurements of real scintillation data, and it has limitations in describing correlated fading processes reflecting actual behavior.

This study proposed a new approach on fading process modeling to better describe the correlated fading processes of real scintillation data. GPS L1/L5 dual-frequency scintillation data collected from Hong Kong was used to characterize correlated fading processes under scintillation. To represent realistic scintillation behavior, the duration for each deep fading event was used to define concurrent deep fades by investigating actual overlaps between deep fades on dual-frequency signals. Also, a new fading process model was constructed to generate fading durations, which helps to relate the characteristics of each deep fading event to the chance of loss-of-lock caused by it. This information helped reduce the conservatism of previous fading process model in which 100% probability of loss-of-lock (given the occurrence of a deep fade) was assumed.

To cover all possible cases of receiver tracking performance, availability simulations were parametrically conducted while varying the probability of loss-of-lock and the reacquisition time of a receiver. LPV-200 availability results under scintillation using the proposed fading process model demonstrated significant availability benefits from dual-frequency SBAS based aviation applications over existing single-frequency SBAS for all values of the two parameters that were varied. This parametric analysis confirmed that aviation availability under scintillation is very sensitive to both of these parameters (probability of loss-of-lock and reacquisition time), which depend on receiver tracking performance. Furthermore, a sensitivity study based on varying GIVEI (for single-frequency SBAS) shows that SBAS ionospheric correction error also has a large impact on aviation availability under scintillation. Because dual-frequency SBAS availability is not dependent on the value of GIVEI that can be supported by single-frequency SBAS, a larger availability benefit is obtained from dual-frequency SBAS when single-frequency SBAS suffers from higher GIVEI.

More accurate availability simulation results can be obtained by specifying receiver tracking parameters and GIVEI values. Receiver tracking performance for a certain deep fading threshold can be used to determine the probability of loss-of-lock and the reacquisition time after loss-of-lock. Hence, further analysis on the relationship between fading characteristics and receiver tracking performance can provide availability-based performance standards for aviation receivers under severe scintillation. Better information regarding SBAS ionospheric correction error values during severe scintillation is also required to obtain more specified availability simulation results.

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